$$\lambda_\gamma \to \frac{1}{16\pi^2}\Big(\frac{2}{5}\overline{C_{\phi2\phi1}}\,g_1^4\,\mathrm{LF}_{3,0}[m_2]+\frac{1}{3}\overline{C_{\phi2\phi1}}\,g_2^4\,\mathrm{LF}_{3,0}[m_2]-\frac{16}{45}\overline{C_{\phi2\phi1}}\,g_2^4\,\mathrm{LF}_{5,-2}[m_2]+$$
$$\frac{1}{36}\overline{C_{\phi2\phi1}}\,g_1^2\left(9\,\overline{y_e}^{pr}y_e^{pr}+4\sum g_2^2\right)\mathrm{LF}_{3,0}[m_{\tilde{l}}^{p}]-\frac{1}{24}\overline{C_{\phi2\phi1}}\,g_1^2\left(6\,\overline{y_e}^{pr}y_e^{pr}+5\sum g_2^2\right)\mathrm{LF}_{4,-1}[m_{\tilde{l}}^{p}]+$$
$$\frac{4}{45}\sum\overline{C_{\phi2\phi1}}\,g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde{l}}^{p}]+\frac{1}{12}\overline{C_{\phi2\phi1}}\,g_1^2\left(9\,\overline{y_e}^{pr}y_e^{pr}-9\,\overline{y_e}^{pr}y_e^{pr}+4\sum g_2^2\right)\mathrm{LF}_{3,0}[m_{\tilde{e}}^{p}]+$$
$$\overline{C_{\phi2\phi1}}\,g_1^2\left(6\,\overline{y_e}^{pr}y_e^{pr}-6\,\overline{y_e}^{pr}y_e^{pr}+5\sum g_2^2\right)\mathrm{LF}_{4,-1}[m_{\tilde{e}}^{p}]+\frac{4}{15}\sum\overline{C_{\phi2\phi1}}\,g_1^4\,\mathrm{LF}_{5,-2}[m_{\tilde{e}}^{p}]+$$
$$\frac{1}{3}\overline{C_{\phi2\phi1}}\,g_1^4\,\mathrm{LF}_{3,0}[\tilde{\mu}]+\frac{1}{6}\overline{C_{\phi2\phi1}}\,g_1^4\,\mathrm{LF}_{4,-1}[\tilde{\mu}]-\frac{4}{15}\overline{C_{\phi2\phi1}}\,g_1^4\,\mathrm{LF}_{5,-2}[\tilde{\mu}]-m_1\tilde{\mu}\,g_1^4\,\mathrm{LF}_{2,2,-1}[m_1,\tilde{\mu}]-$$
$$\overline{C_{\phi2\phi1}}\,g_1^4\,\mathrm{LF}_{2,2,-2}[m_1,\tilde{\mu}]-\frac{1}{12}g_1^4\left(13\,\overline{C_{\phi2\phi1}}+36\,m_1\tilde{\mu}\right)\mathrm{LF}_{3,2,-1}[m_1,\tilde{\mu}]+$$
$$m_1 g_1^4\left(-3\,m_1\overline{C_{\phi2\phi1}}+2\tilde{\mu}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\right)\mathrm{LF}_{2,2,0}[m_1,\tilde{\mu}]+$$
$$\frac{2}{3}\overline{C_{\phi2\phi1}}\,g_1^4\,\mathrm{LF}_{3,1,-1}[m_1,\tilde{\mu}]+\frac{2}{3}\overline{C_{\phi2\phi1}}\,g_1^4\,\mathrm{LF}_{3,2,-1}[m_2,\tilde{\mu}]-$$

[illegible]

$$\frac{1}{2}m_2\tilde{\mu}\,g_2^2\left(3\,g_1^2\left(C_{\phi1^2}+C_{\phi2^2}\right)+2\,g_2^2\left(2\,C_{\phi1^2}+3\,C_{\phi2^2}\right)\right)\mathrm{LF}_{5,1,-2}[\tilde{\mu},m_2]+$$
$$4\,\overline{C_{\phi2\phi1}}\,g_1^2 g_2^2\,\mathrm{LF}_{2,2,1,-2}[m_2,\tilde{\mu},m_1]+$$
$$\frac{1}{4}g_1^2 g_2^2\left(-\tilde{\mu}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\left(7\,m_1+5\,m_2\right)+12\,\overline{C_{\phi2\phi1}}\left(m_1 m_2+\tilde{\mu}^2\right)\right)\mathrm{LF}_{2,2,1,-1}[m_2,\tilde{\mu},m_1]+$$
$$2\,m_1 m_2 C_{\phi2\phi1}\,g_1^2 g_2^2\tilde{\mu}^2\,\mathrm{LF}_{2,2,1,0}[m_2,\tilde{\mu},m_1]-2\,\overline{C_{\phi2\phi1}}\,g_1^2 g_2^2\,\mathrm{LF}_{3,2,1,-3}[m_2,\tilde{\mu},m_1]+$$
$$g_1^2 g_2^2\left(\tilde{\mu}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\left(m_1+m_2\right)-2\,\overline{C_{\phi2\phi1}}\left(m_1 m_2+\tilde{\mu}^2\right)\right)\mathrm{LF}_{3,2,1,-2}[m_2,\tilde{\mu},m_1]-$$
$$2\,m_1 m_2 C_{\phi2\phi1}\,g_1^2 g_2^2\tilde{\mu}^2\,\mathrm{LF}_{3,2,1,-1}[m_2,\tilde{\mu},m_1]-$$
$$\frac{3}{4}\overline{y_d}^{st}y_d^{sr}a_d^{pt}\left(\overline{C_{\phi2\phi1}}\,\overline{a_d}^{pr}+\tilde{\mu}\,\overline{y_d}^{pr}\left(C_{\phi1^2}-C_{\phi2^2}\right)\right)\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{d}}^{t}\right]-$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{d}}^{r},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t}\right]-$$
$$3\,\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{sr}a_d^{pt}\,\mathrm{LF}_{1,1,1,0}\left[m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}\right]+$$
$$\frac{3}{4}\overline{y_d}^{st}y_d^{sr}a_d^{pt}\left(\overline{C_{\phi2\phi1}}\,\overline{a_d}^{pr}+\tilde{\mu}\,\overline{y_d}^{pr}\left(C_{\phi1^2}-C_{\phi2^2}\right)\right)\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r}\right]-$$
$$3\,\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{sr}a_d^{pt}\,\mathrm{LF}_{1,1,1,0}\left[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}\right]+$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{d}}^{t},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r}\right]-$$
$$\frac{1}{4}\overline{y_e}^{st}y_e^{sr}a_e^{pt}\left(\overline{C_{\phi2\phi1}}\,\overline{a_e}^{pr}+\tilde{\mu}\,\overline{y_e}^{pr}\left(C_{\phi1^2}-C_{\phi2^2}\right)\right)\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p},m_{\tilde{e}}^{t}\right]-$$
$$\frac{1}{4}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{e}}^{r},m_{\tilde{l}}^{s},m_{\tilde{e}}^{t}\right]-$$
$$\tilde{\mu}\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{sr}a_e^{pt}\,\mathrm{LF}_{1,1,1,0}\left[m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}\right]+$$
$$\frac{1}{4}\overline{y_e}^{st}y_e^{sr}a_e^{pt}\left(\overline{C_{\phi2\phi1}}\,\overline{a_e}^{pr}+\tilde{\mu}\,\overline{y_e}^{pr}\left(C_{\phi1^2}-C_{\phi2^2}\right)\right)\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{e}}^{t},m_{\tilde{l}}^{p},m_{\tilde{e}}^{r}\right]-$$
$$\tilde{\mu}\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{sr}a_e^{pt}\,\mathrm{LF}_{1,1,1,0}\left[m_{\tilde{e}}^{t},m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}\right]+$$
$$\frac{1}{4}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{e}}^{t},m_{\tilde{l}}^{s},m_{\tilde{e}}^{r}\right]-$$
$$\overline{C_{\phi2\phi1}}\,\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}a_e^{sr}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}\right]+$$
$$\overline{C_{\phi2\phi1}}\,\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}a_e^{sr}\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}\right]-$$
$$\frac{1}{2}\overline{y_e}^{st}y_e^{sr}a_e^{pt}\left(\overline{C_{\phi2\phi1}}\,\overline{a_e}^{pr}+\tilde{\mu}\,\overline{y_e}^{pr}\left(C_{\phi1^2}+C_{\phi2^2}\right)\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{t},m_{\tilde{e}}^{r}\right]+$$
$$\frac{1}{2}\overline{y_e}^{st}y_e^{sr}a_e^{pt}\left(\overline{C_{\phi2\phi1}}\,\overline{a_e}^{pr}+\tilde{\mu}\,\overline{y_e}^{pr}\left(C_{\phi1^2}+C_{\phi2^2}\right)\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{t},m_{\tilde{e}}^{r}\right]-$$
$$\tilde{\mu}\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{sr}\left(C_{\phi2^2}a_e^{pt}+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_e^{pt}\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}\right]+$$
$$\tilde{\mu}\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{sr}\left(C_{\phi2^2}a_e^{pt}+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_e^{pt}\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}\right]+$$
$$\frac{1}{4}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s},m_{\tilde{e}}^{r}\right]+$$
$$\frac{1}{4}\tilde{\mu}\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{sr}a_e^{pt}\left(C_{\phi1^2}-C_{\phi2^2}\right)\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{l}}^{p},m_{\tilde{l}}^{s},m_{\tilde{e}}^{t}\right]-$$
$$\frac{1}{2}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{e}}^{r}\right]+$$
$$\frac{1}{2}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{pt}y_e^{sr}\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{e}}^{r}\right]-$$
$$C_{\phi1^2}\tilde{\mu}\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{sr}a_e^{pt}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}\right]+$$
$$C_{\phi1^2}\tilde{\mu}\,\overline{y_e}^{pr}\,\overline{y_e}^{st}y_e^{sr}a_e^{pt}\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}\right]-$$
$$3\,\overline{C_{\phi2\phi1}}\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_d}^{pr}a_d^{sr}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}\right]+$$
$$3\,\overline{C_{\phi2\phi1}}\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_d}^{pr}a_d^{sr}\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}\right]-$$
$$\frac{3}{2}\overline{y_d}^{st}y_d^{sr}a_d^{pt}\left(\overline{C_{\phi2\phi1}}\,\overline{a_d}^{pr}+\tilde{\mu}\,\overline{y_d}^{pr}\left(C_{\phi1^2}+C_{\phi2^2}\right)\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{d}}^{r}\right]+$$
$$\frac{3}{2}\overline{y_d}^{st}y_d^{sr}a_d^{pt}\left(\overline{C_{\phi2\phi1}}\,\overline{a_d}^{pr}+\tilde{\mu}\,\overline{y_d}^{pr}\left(C_{\phi1^2}+C_{\phi2^2}\right)\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{d}}^{r}\right]-$$
$$3\,\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{sr}\left(C_{\phi2^2}a_d^{pt}+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_d^{pt}\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}\right]+$$
$$3\,\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{sr}\left(C_{\phi2^2}a_d^{pt}+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_d^{pt}\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}\right]+$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\left(\overline{y_u}^{st}y_u^{pt}\,\overline{a_d}^{pr}a_d^{sr}+\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\right)\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r}\right]+$$
$$\frac{3}{4}\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{sr}a_d^{pt}\left(C_{\phi1^2}-C_{\phi2^2}\right)\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t}\right]+$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\left(\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}a_u^{sr}-\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}\right]+$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\left(-\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}a_u^{sr}+\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}\right]+$$
$$\frac{3}{4}\tilde{\mu}\,\overline{y_u}^{st}\left(-\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{pr}y_u^{pt}y_u^{sr}+\overline{y_d}^{pr}y_d^{sr}\left(a_u^{pt}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)+2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{pt}\right)\right)$$
$$\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}\right]+$$
$$\frac{3}{4}\tilde{\mu}\,\overline{y_u}^{st}\left(\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{pr}y_u^{pt}y_u^{sr}-\overline{y_d}^{pr}y_d^{sr}\left(a_u^{pt}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)+2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{pt}\right)\right)$$
$$\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}\right]+$$
$$\frac{3}{8}\overline{C_{\phi2\phi1}}\left(-\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}a_u^{sr}+\tilde{\mu}^2\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{pt}y_u^{sr}\right)\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{q}}^{s}\right]+$$
$$\frac{3}{8}\tilde{\mu}\,\overline{y_u}^{st}\left(\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,\overline{y_u}^{pr}y_u^{pt}y_u^{sr}-\overline{y_d}^{pr}y_d^{sr}\left(a_u^{pt}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)+2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{pt}\right)\right)$$
$$\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s}\right]-\frac{3}{2}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{d}}^{r}\right]+$$
$$\frac{3}{2}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{pt}y_d^{sr}\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{d}}^{r}\right]-$$
$$3\,C_{\phi1^2}\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{sr}a_d^{pt}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}\right]+$$
$$3\,C_{\phi1^2}\tilde{\mu}\,\overline{y_d}^{pr}\,\overline{y_d}^{st}y_d^{sr}a_d^{pt}\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}\right]+$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}a_u^{sr}\,\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}\right]-$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}a_u^{sr}\,\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}\right]+$$
$$\frac{3}{4}\tilde{\mu}\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}\left(a_u^{pt}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)+2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{pt}\right)\mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}\right]-$$
$$\frac{3}{4}\tilde{\mu}\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}\left(a_u^{pt}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)+2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{pt}\right)\mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}\right]-$$
$$\frac{3}{8}\overline{C_{\phi2\phi1}}\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_u}^{pr}a_u^{sr}\,\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{q}}^{p}\right]-$$
$$\frac{3}{8}\tilde{\mu}\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}\left(a_u^{pt}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)+2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{pt}\right)\mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p}\right]-$$
$$\frac{1}{4}g_1^2 g_2^2\left(7\,\overline{C_{\phi2\phi1}}+4\tilde{\mu}\left(m_1+m_2\right)\right)\mathrm{LF}_{2,1,1,-1}[\tilde{\mu},m_1,m_2]+$$
$$\frac{1}{2}m_1 g_1^2 g_2^2\left(-m_2\overline{C_{\phi2\phi1}}+\tilde{\mu}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\right)\mathrm{LF}_{2,1,1,0}[\tilde{\mu},m_1,m_2]+$$
$$3\,\overline{C_{\phi2\phi1}}\,g_1^2 g_2^2\,\mathrm{LF}_{3,1,1,-2}[\tilde{\mu},m_1,m_2]+\frac{1}{8}g_1^2 g_2^2$$
$$\left(-\tilde{\mu}\left(C_{\phi1^2}\left(9\,m_1+5\,m_2\right)+C_{\phi2^2}\left(31\,m_1+19\,m_2\right)\right)+16\,\overline{C_{\phi2\phi1}}\left(m_1 m_2+\tilde{\mu}^2\right)\right)\mathrm{LF}_{3,1,1,-1}[\tilde{\mu},m_1,m_2]+$$
$$m_1 m_2 C_{\phi2\phi1}\,g_1^2 g_2^2\tilde{\mu}^2\,\mathrm{LF}_{3,1,1,0}[\tilde{\mu},m_1,m_2]-\frac{3}{2}\overline{C_{\phi2\phi1}}\,g_1^2 g_2^2\,\mathrm{LF}_{4,1,1,-3}[\tilde{\mu},m_1,m_2]+$$
$$\frac{3}{4}g_1^2 g_2^2\left(\tilde{\mu}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\left(m_1+m_2\right)-2\,\overline{C_{\phi2\phi1}}\left(m_1 m_2+\tilde{\mu}^2\right)\right)\mathrm{LF}_{4,1,1,-2}[\tilde{\mu},m_1,m_2]-$$
$$\frac{3}{2}m_1 m_2 C_{\phi2\phi1}\,g_1^2 g_2^2\tilde{\mu}^2\,\mathrm{LF}_{4,1,1,-1}[\tilde{\mu},m_1,m_2]-2\,\overline{C_{\phi2\phi1}}\,g_1^2 g_2^2\,\mathrm{LF}_{3,2,1,-3}[\tilde{\mu},m_2,m_1]+$$
$$g_1^2 g_2^2\left(\tilde{\mu}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\left(m_1+m_2\right)-2\,\overline{C_{\phi2\phi1}}\left(m_1 m_2+\tilde{\mu}^2\right)\right)\mathrm{LF}_{3,2,1,-2}[\tilde{\mu},m_2,m_1]-$$
$$2\,m_1 m_2 C_{\phi2\phi1}\,g_1^2 g_2^2\tilde{\mu}^2\,\mathrm{LF}_{3,2,1,-1}[\tilde{\mu},m_2,m_1]+$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\,\overline{a_d}^{pr}a_d^{sr}\left(-\overline{a_d}^{st}a_d^{pt}+\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{pt}\right)\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}\right]+$$
$$\frac{3}{4}\overline{a_d}^{pr}a_d^{pt}\left(\overline{C_{\phi2\phi1}}\,\overline{a_d}^{st}a_d^{sr}-\tilde{\mu}\,\overline{y_d}^{st}\left(a_d^{sr}\left(-C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_d^{sr}\right)\right)$$
$$\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^{r},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{q}}^{p}\right]-3\,\tilde{\mu}\,\overline{y_d}^{st}\,\overline{a_d}^{pr}a_d^{pt}a_d^{sr}\,\mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}\right]+$$
$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_d}^{st}a_d^{sr}\left(C_{\phi2\phi1}\,\overline{y_d}^{pr}a_d^{pt}-\overline{C_{\phi2\phi1}}\,y_d^{pt}\,\overline{a_d}^{pr}\right)\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^{t},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}\right]+\frac{3}{4}\tilde{\mu}\,\overline{y_d}^{st}a_d^{pt}$$
$$\left(-C_{\phi2\phi1}\tilde{\mu}\,\overline{y_d}^{pr}a_d^{sr}+\overline{a_d}^{pr}\left(a_d^{sr}\left(-C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_d^{sr}\right)\right)\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{d}}^{t},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}\right]+$$
$$\frac{1}{4}\overline{C_{\phi2\phi1}}\,\overline{a_e}^{pr}a_e^{sr}\left(-\overline{a_e}^{st}a_e^{pt}+\tilde{\mu}^2\,\overline{y_e}^{st}y_e^{pt}\right)\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^{r},m_{\tilde{l}}^{p},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}\right]+$$
$$\frac{1}{4}\overline{a_e}^{pr}a_e^{pt}\left(\overline{C_{\phi2\phi1}}\,\overline{a_e}^{st}a_e^{sr}-\tilde{\mu}\,\overline{y_e}^{st}\left(a_e^{sr}\left(-C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_e^{sr}\right)\right)$$
$$\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^{r},m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{l}}^{p}\right]-\tilde{\mu}\,\overline{y_e}^{st}\,\overline{a_e}^{pr}a_e^{pt}a_e^{sr}\,\mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p},m_{\tilde{l}}^{s}\right]+$$
$$\frac{1}{4}\tilde{\mu}^2\,\overline{y_e}^{st}a_e^{sr}\left(C_{\phi2\phi1}\,\overline{y_e}^{pr}a_e^{pt}-\overline{C_{\phi2\phi1}}\,y_e^{pt}\,\overline{a_e}^{pr}\right)\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^{t},m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}\right]+$$
$$\frac{1}{4}\tilde{\mu}\,\overline{y_e}^{st}a_e^{pt}\left(-C_{\phi2\phi1}\tilde{\mu}\,\overline{y_e}^{pr}a_e^{sr}+\overline{a_e}^{pr}\left(a_e^{sr}\left(-C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_e^{sr}\right)\right)$$
$$\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{e}}^{t},m_{\tilde{l}}^{s},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}\right]-\frac{1}{2}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}a_e^{sr}\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}\right]+$$
$$\frac{1}{2}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_e}^{st}y_e^{pt}\,\overline{a_e}^{pr}a_e^{sr}\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{r},m_{\tilde{e}}^{t},m_{\tilde{l}}^{s}\right]-$$
$$\frac{1}{2}a_e^{pt}a_e^{sr}\left(C_{\phi2\phi1}\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}+\overline{a_e}^{pr}\left(\overline{C_{\phi2\phi1}}\,\overline{a_e}^{st}+2\,C_{\phi2^2}\tilde{\mu}\,\overline{y_e}^{st}\right)\right)\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}\right]+$$
$$\frac{1}{2}a_e^{pt}a_e^{sr}\left(C_{\phi2\phi1}\tilde{\mu}^2\,\overline{y_e}^{pr}\,\overline{y_e}^{st}+\overline{a_e}^{pr}\left(\overline{C_{\phi2\phi1}}\,\overline{a_e}^{st}+2\,C_{\phi2^2}\tilde{\mu}\,\overline{y_e}^{st}\right)\right)\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{l}}^{p},m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{s}\right]-$$
$$\frac{1}{2}\tilde{\mu}\,\overline{y_e}^{st}\,\overline{a_e}^{pr}a_e^{pt}\left(a_e^{sr}\left(C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_e^{sr}\right)\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}\right]+$$
$$\frac{1}{2}\tilde{\mu}\,\overline{y_e}^{st}\,\overline{a_e}^{pr}a_e^{pt}\left(a_e^{sr}\left(C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_e^{sr}\right)\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{l}}^{s},m_{\tilde{e}}^{t},m_{\tilde{e}}^{r},m_{\tilde{l}}^{p}\right]-$$
$$\frac{3}{2}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_d}^{pr}a_d^{sr}\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}\right]+$$
$$\frac{3}{2}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_d}^{st}y_d^{pt}\,\overline{a_d}^{pr}a_d^{sr}\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{d}}^{t},m_{\tilde{q}}^{s}\right]-$$
$$\frac{3}{2}a_d^{pt}a_d^{sr}\left(C_{\phi2\phi1}\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}+\overline{a_d}^{pr}\left(\overline{C_{\phi2\phi1}}\,\overline{a_d}^{st}+2\,C_{\phi2^2}\tilde{\mu}\,\overline{y_d}^{st}\right)\right)\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}\right]+$$
$$\frac{3}{2}a_d^{pt}a_d^{sr}\left(C_{\phi2\phi1}\tilde{\mu}^2\,\overline{y_d}^{pr}\,\overline{y_d}^{st}+\overline{a_d}^{pr}\left(\overline{C_{\phi2\phi1}}\,\overline{a_d}^{st}+2\,C_{\phi2^2}\tilde{\mu}\,\overline{y_d}^{st}\right)\right)\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{s}\right]+$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\left(\tilde{\mu}^4\,\overline{y_d}^{pr}y_d^{sr}\,\overline{y_u}^{st}y_u^{pt}+\overline{a_d}^{pr}\left(a_d^{sr}\,\overline{a_u}^{st}a_u^{pt}+\tilde{\mu}^2\,\overline{y_u}^{st}\left(y_d^{sr}a_u^{pt}+2\,y_u^{pt}a_d^{sr}\right)\right)\right)$$
$$\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}\right]+$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_d}^{st}y_u^{sr}a_d^{pt}\,\overline{a_u}^{pr}\,\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{u}}^{r}\right]-$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_u}^{pr}a_u^{sr}\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}\right]+$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_u}^{pr}a_u^{sr}\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}\right]-$$
$$3\,\tilde{\mu}^3\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}\,\mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}\right]-\frac{3}{4}\tilde{\mu}^2\,\overline{y_u}^{st}$$
$$\left(\overline{C_{\phi2\phi1}}\,y_u^{sr}\,\overline{a_u}^{pr}a_u^{pt}+\overline{y_u}^{pr}\left(2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2 y_u^{pt}y_u^{sr}+a_u^{pt}\left(2\,C_{\phi2\phi1}a_u^{sr}+\tilde{\mu}\,y_u^{sr}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\right)\right)\right)$$
$$\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}\right]+\frac{3}{4}\tilde{\mu}^2\,\overline{y_u}^{st}$$
$$\left(\overline{C_{\phi2\phi1}}\,y_u^{sr}\,\overline{a_u}^{pr}a_u^{pt}+\overline{y_u}^{pr}\left(2\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2 y_u^{pt}y_u^{sr}+a_u^{pt}\left(2\,C_{\phi2\phi1}a_u^{sr}+\tilde{\mu}\,y_u^{sr}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\right)\right)\right)$$
$$\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}\right]+\frac{3}{8}\tilde{\mu}^2\,\overline{y_u}^{st}\left(-\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}\left(y_u^{pt}a_u^{sr}+y_u^{sr}a_u^{pt}\right)+\right.$$
$$\left.\overline{y_u}^{pr}\left(4\,\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2 y_u^{pt}y_u^{sr}+a_u^{pt}\left(4\,C_{\phi2\phi1}a_u^{sr}+\tilde{\mu}\,y_u^{sr}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\right)\right)\right)$$
$$\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}\right]+\frac{3}{8}\tilde{\mu}^2\,\overline{y_u}^{st}\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}\left(y_u^{pt}a_u^{sr}+y_u^{sr}a_u^{pt}\right)-\right.$$
$$\left.\tilde{\mu}\,\overline{y_u}^{pr}y_u^{sr}a_u^{pt}\left(C_{\phi1^2}-5\,C_{\phi2^2}\right)\right)\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}\right]-$$
$$\frac{3}{2}\tilde{\mu}\,\overline{y_d}^{st}\,\overline{a_d}^{pr}a_d^{pt}\left(a_d^{sr}\left(C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_d^{sr}\right)\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}\right]+$$
$$\frac{3}{2}\tilde{\mu}\,\overline{y_d}^{st}\,\overline{a_d}^{pr}a_d^{pt}\left(a_d^{sr}\left(C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_d^{sr}\right)\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{d}}^{t},m_{\tilde{d}}^{r},m_{\tilde{q}}^{p}\right]+$$
$$\frac{3}{4}\tilde{\mu}\,\overline{y_u}^{st}a_d^{sr}\left(\overline{a_d}^{pr}a_u^{pt}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)+\tilde{\mu}\,\overline{y_d}^{pr}\left(2\,C_{\phi2\phi1}a_u^{pt}+\tilde{\mu}\,y_u^{pt}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\right)\right)$$
$$\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{d}}^{r},m_{\tilde{u}}^{t}\right]-\frac{3}{4}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_u}^{pr}a_u^{sr}\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}\right]+$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_u}^{pr}a_u^{sr}\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r},m_{\tilde{u}}^{t}\right]-$$
$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}+\tilde{\mu}\,\overline{y_u}^{pr}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\right)\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}\right]+$$
$$\frac{3}{4}\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{sr}a_u^{pt}\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}+\tilde{\mu}\,\overline{y_u}^{pr}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\right)\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t},m_{\tilde{u}}^{r}\right]+$$
$$\frac{3}{8}\tilde{\mu}^2\,\overline{y_u}^{st}\left(3\,\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}\left(y_u^{pt}a_u^{sr}+y_u^{sr}a_u^{pt}\right)+\tilde{\mu}\,\overline{y_u}^{pr}y_u^{sr}a_u^{pt}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\right)$$
$$\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}\right]+$$
$$\frac{3}{8}\tilde{\mu}^2\,\overline{y_u}^{st}\left(\overline{C_{\phi2\phi1}}\,\overline{a_u}^{pr}\left(y_u^{pt}a_u^{sr}+y_u^{sr}a_u^{pt}\right)+\tilde{\mu}\,\overline{y_u}^{pr}y_u^{sr}a_u^{pt}\left(3\,C_{\phi1^2}+C_{\phi2^2}\right)\right)$$
$$\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{s},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{u}}^{r}\right]-$$
$$\frac{3}{2}\tilde{\mu}^2\,\overline{y_u}^{st}a_u^{pt}\left(\overline{C_{\phi2\phi1}}\,y_u^{sr}\,\overline{a_u}^{pr}+\overline{y_u}^{pr}\left(C_{\phi2\phi1}a_u^{sr}+2\,C_{\phi2^2}\tilde{\mu}\,y_u^{sr}\right)\right)\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}\right]+$$
$$\frac{3}{2}\tilde{\mu}^2\,\overline{y_u}^{st}a_u^{pt}\left(\overline{C_{\phi2\phi1}}\,y_u^{sr}\,\overline{a_u}^{pr}+\overline{y_u}^{pr}\left(C_{\phi2\phi1}a_u^{sr}+2\,C_{\phi2^2}\tilde{\mu}\,y_u^{sr}\right)\right)\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{t}\right]-$$
$$\frac{3}{2}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_u}^{pr}a_u^{sr}\,\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}\right]+$$
$$\frac{3}{2}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{pt}\,\overline{a_u}^{pr}a_u^{sr}\,\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{u}}^{r},m_{\tilde{q}}^{s},m_{\tilde{q}}^{p},m_{\tilde{u}}^{t}\right]+$$
$$\frac{3}{4}\overline{C_{\phi2\phi1}}\,\tilde{\mu}^2\,\overline{y_u}^{st}y_u^{pt}\left(\overline{a_u}^{pr}a_u^{sr}+\tilde{\mu}^2\,\overline{y_u}^{pr}y_u^{sr}\right)\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{u}}^{r},m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}\right]-$$
$$\frac{3}{2}\tilde{\mu}^3\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}\left(a_u^{pt}\left(C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{pt}\right)\mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}\right]+$$
$$\frac{3}{2}\tilde{\mu}^3\,\overline{y_u}^{pr}\,\overline{y_u}^{st}y_u^{sr}\left(a_u^{pt}\left(C_{\phi1^2}+C_{\phi2^2}\right)+\overline{C_{\phi2\phi1}}\,\tilde{\mu}\,y_u^{pt}\right)\mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{u}}^{t},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s},m_{\tilde{u}}^{r}\right]+\frac{3}{4}\tilde{\mu}^2\,\overline{y_u}^{st}$$
$$a_u^{pt}\left(\overline{C_{\phi2\phi1}}\,y_u^{sr}\,\overline{a_u}^{pr}+\overline{y_u}^{pr}\left(C_{\phi2\phi1}a_u^{sr}+\tilde{\mu}\,y_u^{sr}\left(C_{\phi1^2}+3\,C_{\phi2^2}\right)\right)\right)\mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{u}}^{t},m_{\tilde{u}}^{r},m_{\tilde{q}}^{p},m_{\tilde{q}}^{s}\right]\Big)$$